



Faculty of Technology and Engineering
Chandubhai S. Patel Institute of Technology
Department of Computer Science & Engineering

Student ID:

Date: / /

Practical Performa

Academic Year	:	2025-26	Semester	:	3 rd
Course code	:	CSUC201	Course name	:	Fundamentals of Data Structure and Algorithms

Practical- 9 Graphs

Questions and Answers (Handwritten):

9.1 Fill in the Blanks (Graph Traversals)

1. BFS (Breadth-First Search) uses a _____ (Queue/Stack) data structure.
2. DFS (Depth-First Search) can be implemented using _____ (Stack/Queue).
3. Time complexity of BFS/DFS for a graph with V vertices and E edges is O(_____).
4. In BFS, nodes are explored in _____ order, while in DFS, nodes are explored in _____ order.

9.2 Programming/Algorithmic Question (Traversal)

Write an algorithm (pseudocode or in any language) to perform **BFS traversal** of a graph using **Adjacency List representation**.

Input Example:

Vertices: {A, B, C, D, E}

Edges: { (A,B), (A,C), (B,D), (C,E) }

Expected Output (starting from A):

A B C D E

9.3 Analytical Question (Topological Sort)

A project has the following dependencies:

- Task A must finish before Task C and Task D.
- Task B must finish before Task D.
- Task C must finish before Task E.
- Task D must finish before Task E and Task F.

- (a) Represent this as a **Directed Acyclic Graph (DAG)**.
- (b) Perform **Topological Sort** and list one possible valid ordering of tasks.
- (c) If an edge is added from **E** $\rightarrow$ **A**, explain what happens and why.

9.4 Application-Oriented Question (Graph Usage)

Consider the following real-world scenarios. For each, identify:

1. Whether it can be modeled as a **directed or undirected graph**.
2. Which graph algorithm is most appropriate.

- (a) **Social network friendship connections** (suggest friends, find communities).
- (b) **Navigation system (Google Maps)** to find the shortest route between two cities.
- (c) **Course prerequisite system** in a university.
- (d) **Network packet routing** where cycles may occur but shortest delay path is needed.

Grade/Marks
Date

(____ / 10)

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